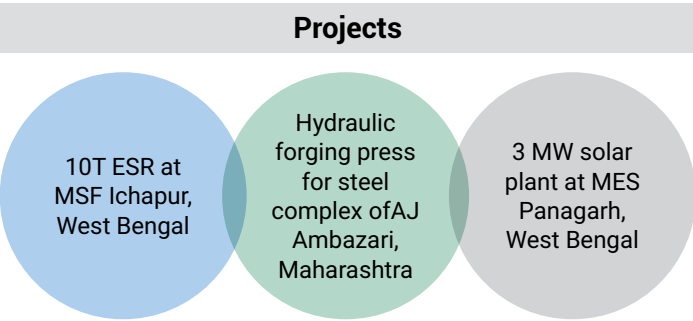


Our business:

Strategic Business Division

Overview

Our solutions include coal washing plants, ore beneficiation plants, utilities and balance of steel plants, industrial furnaces for heating and melting, and electrostatic precipitators.



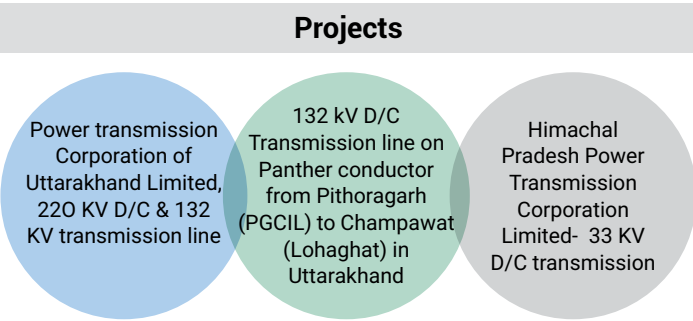
Hydraulic press for OFAJ, Ambazari, Maharashtra

Our business:

Power Transmission Division

Overview

This division executes EPC projects for transmission towers to address the rising demand for new power evacuation lines. It serves PGCIL (Power Grid Corporation of India Limited), State Electricity Boards, and private players like Tata, L&T, and Sterlite, covering lines from 132 KV to 1100 KV and associated substations.



Power transmission project, Uttarakhand

Our business:

Water Management Division



Overview

The water management division of BTL EPC delivers comprehensive solutions for water and wastewater treatment. This division plays a key role in executing standalone water projects and integrating water systems within larger industrial and infrastructure EPC contracts. It serves public and private sector clients, ensuring a compliance with environmental norms and promoting sustainable practices. Key clients include Jal Jeevan Mission in Assam, Jharkhand Urban Infrastructure Development Company Limited, NTPC Talcher and DVC, Bokaro. These solutions include coal washing plants, ore beneficiation plants, utilities and balance-of-steel plants, industrial furnaces for heating and melting, and electrostatic precipitators. The water management division is crucial to BTL EPC's growth strategy, contributing to revenue diversification, enhancing its ESG performance, and addressing the growing need for efficient water management across industries.

Policy

The water management division of BTL EPC operates under a sustainability-driven policy framework that prioritises water reuse, pollution control, and strict compliance with environmental regulations. Reinforcing this approach, the Government of India, through the Ministry of Jal Shakti has introduced national initiatives such as the Jal Shakti, Abhiyan: Catch the Rain 2024–25 campaign, themed Nari Shakti se Jal Shakti. This program promotes women-led water conservation efforts through rainwater harvesting, groundwater recharge, watershed development, and afforestation.

Growth

The water management division at BTL EPC witnessed early-stage traction in FY 2024–25. The division demonstrated resilience by mobilising new resources, onboarding contractors, and stabilising project delivery. These efforts have laid the groundwork for growth.

Strengths

BTL EPC's involvement in the water management division is supported by key internal strengths, such as:

Resource identification and recruitment: The business successfully identified the resource gaps and initiated recruitment, onboarding mostly new personnel to support ongoing and upcoming projects.

In-house functional control: While external resources were engaged due to limitations in internal capacity, the Company ensured total in-house control over critical functions such as engineering, procurement, and related project support domains.

Highlights, FY 2024-25

- Commissioning of the Deoghar Septage Treatment Plant marked a milestone in municipal waste management.
- Completion of the Hazaribagh Septage Plant highlighted execution capability despite challenges.
- Secured the first major wastewater treatment plant order, a landmark 2,000 cum/hour capacity project from BHEL, Talcher.
- 80% of job for ETP , DVC is complete.

Challenges and mitigations, FY 25

Challenges:

Team formation issues: The segment lacked an assigned and structured team.

Execution risks: Two projects were in critical condition, with multiple execution challenges.

JJM project under performance: The JJM initiative was facing delays and payment issues.

Mitigation measures:

Team mobilisation: The Company supported the formation of a dedicated team, partially resolving the resource gap.

Operational autonomy: Teams were empowered to engage directly with clients and consultants to stabilise project execution and transition into the O&M phase.

Engineering troubleshooting: A new establishment was on-boarded. Joint reviews with clients helped to address fundamental engineering challenges.

Outlook, FY 2025-26

The Company intends to build on its strengths in FY 2025–26 by tapping into significant opportunities available in the water and wastewater segment, particularly within power plants operated by BHEL, NTPC, DVC, and other private sector players. To pursue this growth, the division requires better structuring and strengthening through the recruitment of skilled professionals and the establishment of a robust organisational framework capable of handling future business growth.





Our business: **Underground Mining**

Overview

Underground conveyor systems are critical in mines due to their ability to transport bulk materials efficiently, safely, and economically especially in deep or complex mining operations.

Unlike trucks, conveyors provide non-stop, automated material movement, significantly increasing throughput. Conveyors can handle long and complex routes, even in deep underground mines where trucks would be inefficient or impractical. They can transport thousands of tons per hour—ideal for modern high-output mines. They eliminate diesel-powered equipment for haulage, leading to reduced ventilation needs and improved air quality. Fewer operators are needed for haulage, reducing human presence in hazardous areas. Conveyors consume less energy per ton of material moved compared to trucks, especially over long distances. Because conveyors do not emit

exhaust fumes, ventilation energy demand drops, especially critical in deep mines. Electrified systems are cleaner and align with decarbonisation goals.

Underground conveyor systems are not just a technical option; they are a strategic necessity in modern mining for enhancing productivity, safety, environmental compliance, and cost efficiency. As mines go deeper and ESG pressures mount, conveyors become the backbone of sustainable underground operations.

The manufacturing division at BTL EPC is a leading player in providing specialised solutions for coal mine conveyor applications. We design and produce dual-drive conveyors/drive heads in various power combinations, ranging from 2 x 37 KW to an industry-leading 2 x 600 KW, marking the highest installed motor power in India.

Over the last 30 years, we have supplied our own make drive heads and conveyors to all coal india and its subsidiaries as well as private mines.

Collieries	Total no. of conveyors
South Eastern Coalfields (SECL)	250
Mahanadi Coalfields (MCL)	21
Western Coalfields (WCL)	57
Bharat Coking Coal Limited (BCCL)	14
Eastern Coal Fields (ECL)	45
Singareni Collieries (SCCL)	55
Central Coalfields (CCL)	1
Private Collieries	25

Underground conveyors and Drive heads supplied

INDU	2x600 KW conveyors supplied
Ultratech	2x315 KW conveyors supplied
JMS Mining	2x250 KW conveyors supplied

Scalability and capacity enhancement

Then: Low-capacity belt conveyors (<500 TPH) with limited lengths (hundreds of meters). Used mostly in shallow or moderately deep mines with low tonnage.

Now: High-capacity systems handling 5,000+ TPH over several kilometers underground. Modular designs allow conveyor extensions as the mine face advances. Used in longwall, bord and pillar, and block caving systems for continuous haulage.

Enhanced safety features

Then: Basic pull cords and trip switches, mostly manual detection of failures. Frequent belt fires due to slippage, overheating, or friction.

Now: Embedded safety features, including Belt drift detectors, Rip

detection sensors, belt misalignment and tear detectors, temperature and fire detection systems, automated emergency shutoff systems and zone-wise fault isolation, Integration with mine-wide emergency systems for alerts and ventilation control.

Reduced maintenance and self-diagnostics

Then: All modifications in conveyor belt systems was done when the system reported downtime.

Now: Predictive maintenance uses vibration sensors on pulleys and gearboxes. Thermal sensors are used in bearings and motors. AI-based

analytics is used in wear prediction. Remote diagnostics platforms enable off-site monitoring and fault troubleshooting. Improved idler design (low-friction, sealed-for-life rollers) for longer life and reduced noise.

Digital integration

Then: The integration with other systems was done manually through human intervention.

Now: Modern underground conveyors are connected to fleet management systems, production scheduling

platforms and ventilation-on-demand systems (which adjust airflow based on belt operation). This enables fully integrated underground-to-surface material flow, often monitored from a remote-control center.

Stronger and smarter belt technologies

Use of steel cord belts instead of fabric belts. Higher tensile strength (up to 8,000 N/mm or more). Lower elongation and longer lifespan. Flame-resistant, anti-static (FRAS)

belts are now standard for safety in gassy mines. Sensor-embedded belts can monitor wear, slippage, and tension in real-time.

Automation and monitoring systems

Integration of PLC-SCADA systems with underground conveyors: Real-time monitoring of belt speed, load, temperature, and motor condition; auto-start/stop sequencing based

on production demand. Advanced control algorithms for synchronised multiple conveyor drives and load sharing.

Energy efficiency and regenerative drives

Use of variable frequency drives (VFDs) on motors to optimise energy use. Regenerative braking systems recover energy on downhill conveyors

and feed it back to the grid or mine power system. Soft starters and load-sharing logic reduce mechanical stress and power surges.

Our business: Agrimech Division



7P3 Petrol Power Weeder

Overview

BTL EPC’s Agrimech division has been committed to advance farm mechanisation in India for over three decades, with a focus on the needs of small and marginal farmers. The division offers a range of quality, Made-in-India agricultural equipment aimed at improving farm productivity. Backed by a strong network of seven branch offices and over 500 dealers across the country, the division ensures easy accessibility, prompt service, and real-time support through trained technicians and a dedicated customer care centre.

Policy

The Sub-Mission on Agricultural Mechanization (SMAM) is a centrally sponsored scheme under the Department of Agriculture and Farmers Welfare, aimed at enhancing farm mechanisation, particularly among small and

marginal farmers. The scheme offers substantial subsidies ranging from 40% to 80% on the purchase of agricultural machinery such as tractors, tillers, FMB (Farm machinery bank) - one line, FSSM (Financial support scheme for farm mechanization) and harvesters, making mechanisation more affordable with support of all states’ government. SMAM supports the establishment of Custom Hiring Centres and machinery banks, which facilitate the rental of farm equipment in regions with low levels of mechanisation, thereby improving accessibility and promoting inclusive agricultural growth.

Growth

The Agrimech division recorded a robust 32% year-on-year revenue growth in FY 2024-25. This strong performance reflects increased market demand and improved sales execution.

Strengths

- Strong leadership and strategic guidance from the senior management.
- Clear vision and execution from a committed leadership team.

Highlights, FY 2024-25

- Achieved significant year-over-year revenue growth of 32%.
- **Total sales:** Over 8,000 weeders and 1,850 power tillers were sold.

Challenges and mitigations, FY 25

- The division underwent a leadership transition with a change in the senior management.
- Employee retention and morale.
- Operational inefficiencies in supply chain and fund flow management.

Mitigation:

- The Company undertook team restructuring through a strategic realignment and strengthened its leadership framework.
- Implemented employee recognition initiatives to enhance motivation and reduce attrition.
- Implemented operational improvements, including better inventory management, optimised handling of debtors and creditors, and streamlined fund flows with improved cash planning.

Outlook, FY 25-26

The Company aims to build on its strengths by continuing to invest in its workforce through structured recognition programs and capability development initiatives.

The division plans to expand its presence in emerging and underpenetrated markets by strengthening dealer networks and actively participating in government tenders. Efforts will be made to digitalise operations for better forecasting, enhanced supply chain visibility, and improved financial planning. After-sales service will be reinforced to boost customer satisfaction and brand loyalty. To sustain the momentum, the Company will introduce new models around regional needs.



Virat 13HP Power Tiller

Our business: **Manufacturing Division**



State-of-the-art manufacturing excellence

State-of-the-art manufacturing facilities are spread over more than 10,00,000 square feet in Kolkata, Kharagpur, and Durgapur comprising modern equipment and skilled workforce supported by a large vendor base.

The Company developed versatile manufacturing capabilities by producing a spectrum of heavy engineering components, including the following:

- Belt conveyor systems and components
- Equipment for bulk material handling projects

- Ash Handling Equipment
- Pressure Vessels
- Ducts
- Tanks
- Refinery columns
- Steel silos

The Company leverages cost-efficient in-house manufacturing facilities to support EPC operations, providing a competitive edge over peers reliant on external procurement.

The Company possesses a strong production capability with over 24,000 MT/year fabrication capacity, along

with 1,80,000 rollers and 1800 pulley production / year, enabling the on-time execution of large EPC projects.

The Company is equipped with advanced in-house facilities for NDT, hydro testing, radiography, and painting, ensuring an adherence with global quality benchmarks.

The Company provides end-to-end fabrication solutions, covering the full cycle from material handling to dispatch, backed by in-house design and engineering competencies.

The Company's commitment to manufacturing excellence is underpinned by state-of-the-art infrastructure, equipped with the latest CNC machines. This allows us to produce not only complex industrial products but also ensures timely delivery with rigorous adherence to quality and industry standards.

Major machines like bearing pushing machine , Pipe boring, Welding, Pipe Cutting, ESAB CNC Plate cutting

Machine, Paint Booth, Shielded Metal Arc Welding, Submerged-Arc Welding, Gas Metal Arc Welding, Gas Tungsten Arc Welding, CNC profile cutting, CNC turning center and others.

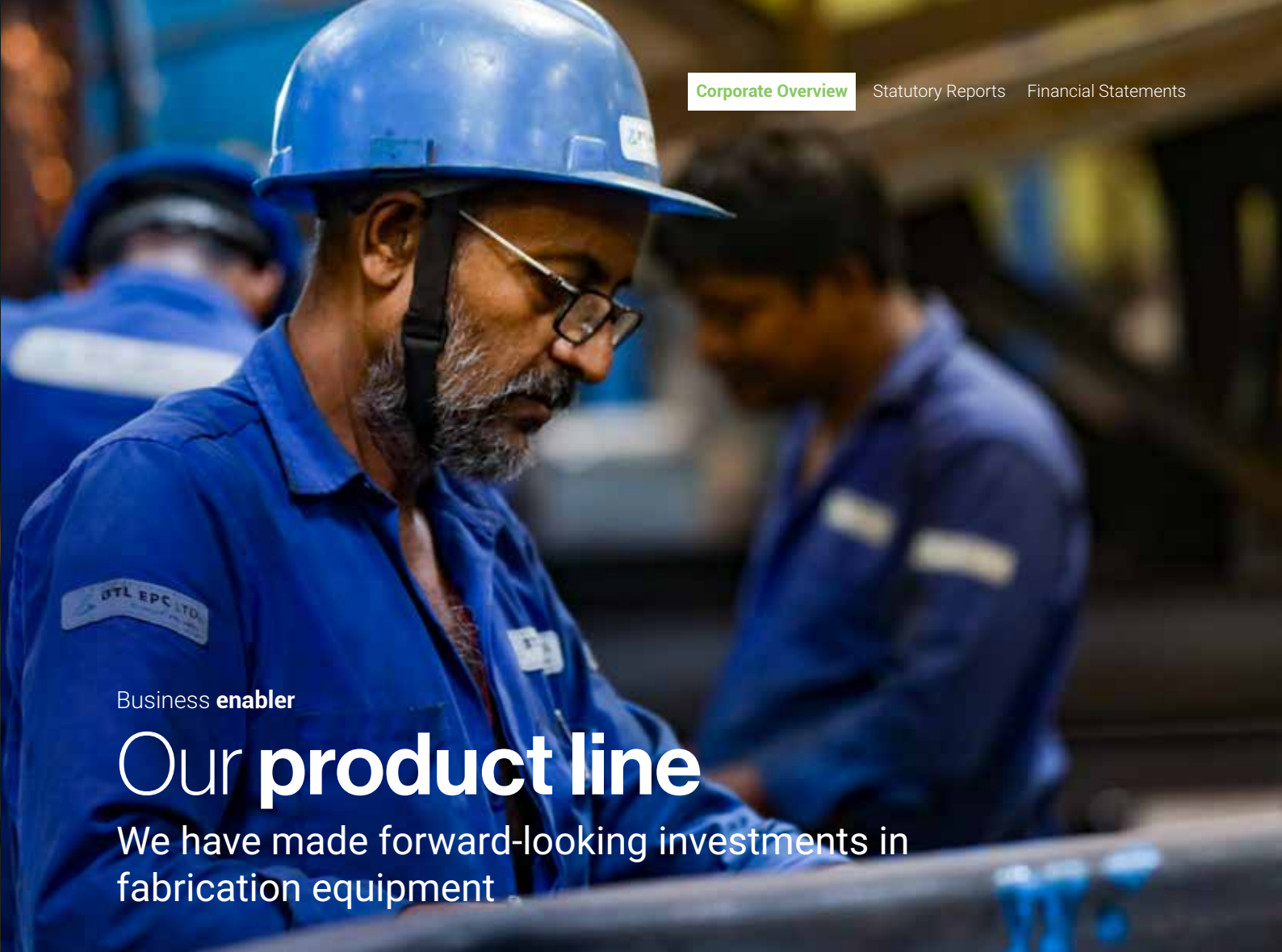
Policy

The Indian government prioritises domestic manufacturing, particularly through initiatives like 'Make in India' and Production-Linked Incentive (PLI) schemes. These policies aim

to transform India into a global manufacturing hub by encouraging domestic and foreign investment, fostering innovation, and developing world-class infrastructure.

Our manufacturing strengths and competencies

We have created an innovation-led culture in heavy engineering solutions



Business enabler

Our product line

We have made forward-looking investments in fabrication equipment

Core skills: Heavy engineering fabrication competence (large height and weight).

Specialisation: Established competence in the fabrication of complex pressure vessels.

Infrastructure: Aggregate 6,96,020 sq ft fabrication space across five units.

Quality: All fabricated material is radiography-tested resulting in a first-time right culture.

Approval: All fabricated material is approved by the Indian Boiler Association.

Value chain: Integrated competence from designing equipment to fabrication.

Repertoire: Diverse mechanical capabilities (fabrication, machining, welding, blasting and painting).



Idlers and pulleys: Essential components for belt conveyors; ensure smooth movement, load support, and belt tracking. Available in carrying, return, and impact types.

Gates (Rod Gate, Slide Gate, Rack and Pinion Gate): Control devices for regulating or diverting bulk material flow in chutes and hoppers.

Chutes: Customised transfer points designed to minimise spillage and wear.

Conveyor structures: Supporting frames for belt conveyors, engineered for stability and durability.

Shuttle conveyors: Mobile conveyors that move back and forth to distribute material over a wider area.

Scrapers/Ploughs: Installed for effective belt cleaning and minimising spillage.

Tripper: Used for controlled discharge of bulk materials along conveyor belts in material handling systems.

Flap gate: A flow control device used in ducts and chutes to regulate or isolate the passage of materials.

Silo: A vertical storage structure designed for holding bulk materials like cement, ash, or grains.

Paddle feeder: A mechanical device that extracts and feeds bulk materials uniformly from storage hoppers or bunkers.

Shell and tube heat exchanger: A thermal device used to transfer heat between two fluids through multiple tubes housed within a shell.

Primary gas cooler: A heat exchanger that reduces the temperature of hot gases in process systems before further handling.



Gear box



Bracket



Steam generator (kettle type)



Pulley



Feeder

Pressure vessel: A container engineered to hold gases or liquids at high pressures safely and efficiently.

Drive head: A critical conveyor component that houses the motor and gearbox to power belt movement.

Underground conveyor gear box: A specially designed gearbox for underground conveyors, enabling reliable power transmission in confined spaces.

Tanks: Engineered containers used for storing liquids, chemicals, or slurries across industrial processes.

Underground mining conveyors: Heavy-duty conveyor systems designed for coal and mineral transport in underground mines. Features include flame-proof drives, dust-tight construction, and high load capacity.

Stainless steel process columns: Vertical process vessels used for distillation, absorption, or chemical separation. Designed in SS material for corrosion resistance, high purity, and chemical compatibility.

Stacks: Tall chimneys or exhaust structures designed for emission dispersion. Engineered for structural stability, corrosion resistance, and environmental compliance.

Tar decanters: Equipment for separating tar from water and sludge in coal chemical plants. Essential for downstream tar distillation and recovery operations.

Ash handling equipment: Systems for collection, conveying, storage, and disposal of bottom ash and fly ash from power plants. Includes submerged scrapers, pneumatic conveying systems, tanks and silos.

6 ways we have transformed in 10 years

1 Plate cutting transition from manual to CNC

Reality: This was manual process required around 20 workers to cut 400 tonnes of plate per month. This affected productivity and consistency.

Initiative: The Company began transitioning to CNC-based plate cutting a decade ago when it procured its first CNC machine (second in 2023, and third in 2025).

Capabilities: The MAKE-ESAB machines comprise oxy-fuel and plasma cutting systems (60mm piercing / 200mm cutting

for oxy-fuel and 32mm piercing / 60mm cutting for plasma).

Improvement: Productivity has improved. The number of workers required to cut 400 tonnes per month has decline from 20 to 6; along with enhancements in cutting quality, cost efficiency, and delivery timelines.

Outcome: The Company intends to increase automated plate cutting from 30% to 90%.

2 Painting of rollers

Reality: Roller painting was carried out by placing them vertically on the shop floor. This method allowed 150–200 rollers per batch with drawbacks (inconsistent thickness, top end receiving less paint). The absence of a closed booth system meant painting could only be done during non-production hours from 10 PM to 6 AM, a bottleneck.

Initiative: An enclosed paint booth chamber was introduced composing a front draft cyclone system to control and extract paint fumes, ensuring better safety and environmental health conditions.

Improvements: The new system increased roller accommodation 300% within the same footprint (450 rollers per batch). Continuous painting across 24 hours eliminated time-based restrictions. Paint consumption declined 20%, while quality and throughput improved.

Outcomes: The new setup allows round-the-clock painting, enhances worker safety, and improves paint application consistency. It also includes a three-tier roller arrangement, enhancing space utilisation.

3 Process mapping

Reality: There was no formal process mapping in the manufacturing unit. Production flow and workstation performance were managed without structured analysis. This made it difficult to identify inefficiencies, delays, and optimisation areas.

Initiative: The Company introduced advanced industrial engineering practices (time-motion studies and value stream mapping). These methodologies helped identify bottlenecks in workstations,

analyse workflows, and map each stage of the production process for rollers and pulleys.

Improvement: Roller production capacity increased from 8,000 to 12,000 units per month; pulley production capacity improved from 110 to 150 units per month.

Outcome: This highlighted the value of data-driven process analysis and in workflow management improvements.

4 Kaizen

Reality: There was no structured continuous improvement method in place.

Initiative: The Kaizen initiative was introduced in September 2022, involving all engineers and workers collectively (rather than through teams). The 5S approach was implemented, with responsibilities assigned. Reviews were conducted

regularly; engineers and workers were recognised every six months.

Improvement: The initiative fostered total employee involvement, resulting in noticeable improvements in workplace safety and housekeeping standards.

Outcome: Worker morale improved.

5 Manufacturing capacity

Reality: In FY 2020–21, the Company operated with limited production capacity - in-house manufacturing capacity of approximately 5,000 tonnes per year and no outsourcing.

Initiative: To enhance output, the Company began to outsource manufacturing. As part of this initiative, the Company engaged

renior consulting for a manufacturing excellence exercise.

Improvement: The Company scaled capabilities - in-factory production reached 8,000–10,000 tonnes per year and outsourced production contributes an additional 10,000–15,000 tonnes a year a near-500% increase in total capacity.

6 Production capacity

Reality: In 2019–20, the Company's production capacity for key components was constrained with rollers at 8,000 numbers /month, pulleys at 110 numbers /month, gear boxes at 20 numbers /year, travelling trippers at 10 numbers /year, and structural items at 3,000 tonnes/year.

Initiative: The Company undertook a combination of outsourcing and capacity expansion, including the addition of new fabrication works and vendor development.

Improvement: By 2024–25, production capacity reported a marked increase rollers

scaled to 12,000 numbers/month, pulleys at 150 numbers/month, gear boxes at 40 numbers/year, and travelling trippers at 40 numbers/year through outsourcing. Structural fabrication capacity grew significantly from 3,000 tonnes/year to 10,000 tonnes/year through in-house enhancements and outsourced support.

Outcome: The Company is better equipped than ever to respond to the anytime and any quantity fabrication needs of large and prominent customers.



Our milestones

We have grown across the years by witnessing in our capabilities and infrastructure, emerging as a responsible heavy engineering company

1965-1970: Established a manufacturing plant in partnership with UK's Tempered Tools Ltd., becoming a key supplier to steel plants, paper mills, plywood factories, and rail factories across India.

1970: Partnered DRDO to develop indigenous defence equipment and secured crucial quality certifications (DQAE and DQAV). Expanded the fabrication unit to produce boiler accessories, including air preheating systems, enhancing capabilities to serve defence and core infrastructure projects.

1976: Pioneered the indigenisation of border fencing with the 'punched tape concertina coil,' developed in collaboration with DRDO.

1977-1990: Emerged as a key supplier of mining conveyors and spares for Coal India. Diversified beyond traditional industries, producing a range of equipment from mechanical mine line equipment to mobile water purification plant to rocket launching pads.

1998: Acquired APV Engineering Company, gaining valuable expertise as a dairy plant and food processing EPC. Completed the Koraput dairy plant in 2000 with a milk processing capacity of 2,00,000 litres per day.

2000: Boosted efficiency by acquiring an adjacent factory to establish a fully automated conveyor roller and pulley unit.

2001: Registered with Engineers India Limited which enabled the direct supply of pressure vessels and heat exchangers to petrochemical and refinery giants, independently and through multinational EPC contractors. Secured major projects with Haldia Petrochemicals, IOCL-Barauni, and IOCL-Haldia.

2002: Expanded its production capacity by acquiring a Durgapur factory, enhancing our fabrication capabilities for larger projects.

2005: Our track record of supplying to Coal India enabled us to win major material handling projects, achieving success with clients such as BPSCL, CESC, ECL, and NTPC. Leveraged our pressure vessel expertise to enter

Coal Chemical By-Product projects in steel plants. Collaborated with VUKhIN Russia and executed projects like Durgapur BOF Gasholder, RSP Gas Condensation Plant and Bokaro Steel Plant Decanter.

2012: Morphed into a full-fledged industrial solutions provider, from underground mining conveyors to complex material handling projects for diverse industries. This diversification extended our annual turnover past ₹200 Crore.

2013: BTL transformed into BTL EPC Ltd. This new name reflected our position as a complete EPC contractor, reflecting our diverse projects across industries.

2014: Strategically ventured into Power Transmission EPC projects for state electricity companies, securing three initial projects and qualifying for high-voltage 400 kV power projects.

2018: Expanded into the ordinance factory sector, supplying and commissioning specialised machines for defence needs. Delivered a 10-ton ESR plant project and secured additional orders via joint ventures.

2019-2021: Despite the global pandemic's impact on industries, BTL EPC Ltd. maintained operations. Our resilience, coupled with the gap left by struggling competitors, led to new material handling and CCD projects. We completed a 3 MW solar power

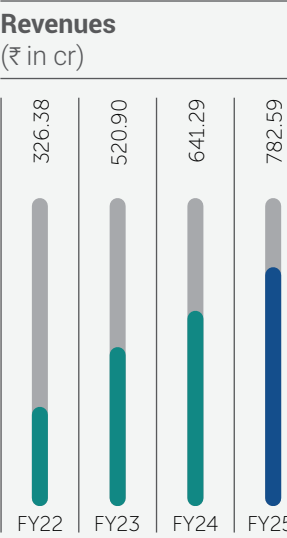
project for the Ministry of Defence in 2020.

2022: Entered ash handling projects for thermal power plants, securing India's largest project through successful international collaboration.

2023 and beyond: Turnover surged to ₹550 Crore. Our strong foundation in manufacturing, engineering, and project execution, along with a ₹2,000 Crore order book, positions us for continued success. We plan to address EPC projects in the water division for STP (Septage Treatment Plant), WTP (Water Treatment Plant) and water distribution; special businesses like mineral beneficiation, FGV and ordinance factory machinery projects through foreign collaborations.

How we have grown over the years

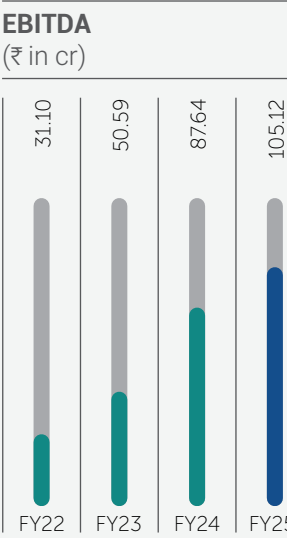
We have responded favourably to the Indian economic growth story by reporting an increased profitability coupled with a decline in borrowings



Meaning
Revenue is the total income generated from the sale of goods related to the Company's operations.

Importance
Revenue represents a basis for the comparison of the Company's competence with competitors.

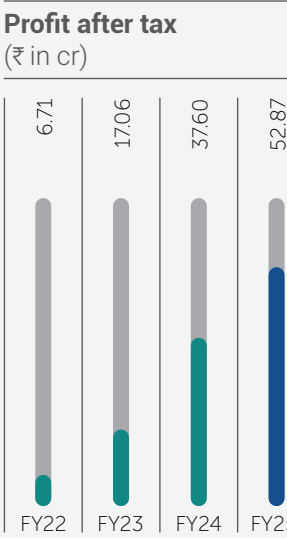
Performance
The Company reported ₹782.59 Crore of revenue in FY 2024-25, 22% higher than the previous year.



Meaning
EBITDA is an acronym that refers to the earnings of a company before interest, tax, depreciation and amortisation expenses are deducted or provided for.

Importance
Investors use EBITDA to assess a company's profitability and efficiency and compare with similar businesses.

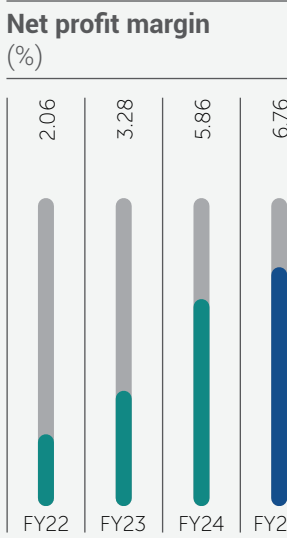
Performance
The Company reported ₹105.12 Crore EBITDA in FY 2024-25, 21% higher than the previous year.



Meaning
The amount left after a company has paid all its operating and non-operating expenditures, other liabilities, and taxes is known as profit after tax. This profit is either allocated to the entity's shareholders as dividend or held in reserve as retained profit.

Importance
Profit after tax represents the actual profit a company earns in a financial year. It reflects the cost structure, business model, and overall competitiveness.

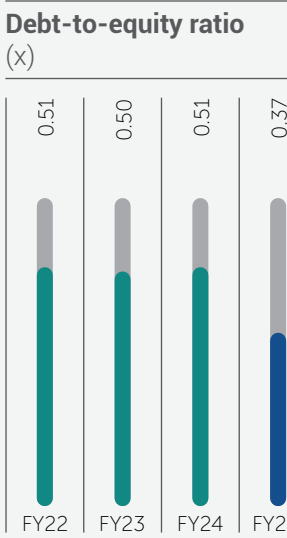
Performance
The Company reported ₹52.87 Crore of PAT in FY 2024-25, 41% higher than the previous year, indicating that its growth had been profitable.



Meaning
Net profit margin (also known as 'profit margin' or 'net profit margin ratio') is a financial measure to calculate the percentage of profit a company produces from its total revenue.

Importance
It represents the net profit that a company earns for each rupee of revenue, expressed as a percentage.

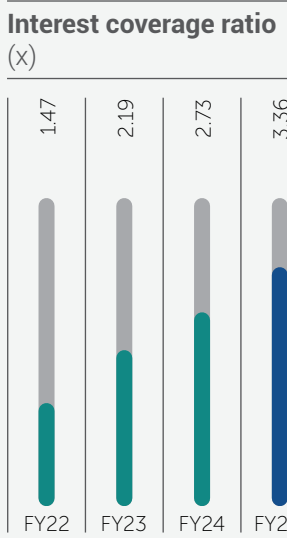
Performance
The Company reported 6.76% net profit margin in FY 2024-25, 90 basis points higher than the previous year.



Meaning
The debt-to-equity ratio (also called the 'debt-equity ratio', 'risk ratio', or 'gearing'), is a leverage ratio that calculates the weight of total debt and financial liabilities against total shareholders' equity.

Importance
This ratio indicates whether a company's capital structure leans more toward debt or equity financing.

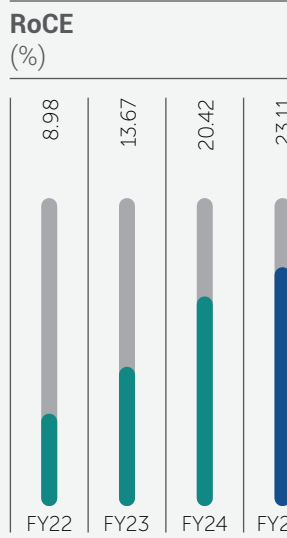
Performance
The Company reported a 0.37 debt-to-equity ratio at the close of FY 2024-25, indicating the under-borrowed nature of the business.



Meaning
The interest coverage ratio is a financial ratio to determine how comfortably a Company can pay interest on its outstanding debt.

Importance
Lenders, creditors and investors commonly use the interest coverage ratio to assess the risk of lending to the Company.

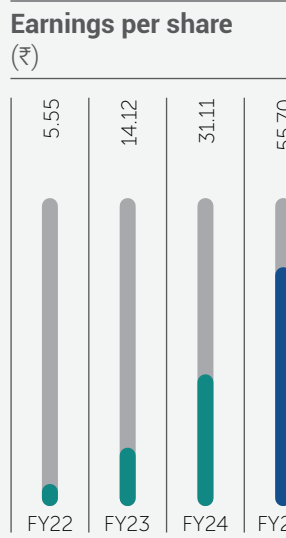
Performance
The Company reported a safe and attractive interest coverage ratio of 3.38x for FY 2024-25, which is improved by 0.63x.



Meaning
Return on capital employed (ROCE), a profitability ratio, measures how efficiently a company is using its capital to generate profits.

Importance
The return on capital employed (ROCE) is one of the most important profitability ratios, commonly used by investors to assess whether a company represents a sound investment.

Performance
The Company reported 23.11% return on capital employed for FY 2024-25, which was 269 bps higher than the previous year on account of larger contracts, upgraded manufacturing, enhanced production and disciplined working capital management.



Meaning
Earnings per share (EPS) is a financial metric, which divides net earnings available to common shareholders by the average outstanding shares for a certain period.

Importance
Earnings per share reflect a company's ability to generate net profits for common shareholders.

Performance
The Company reported ₹55.70 earnings per share for FY 2024-25, which was 79% higher than previous year.

PART TWO

Building for India. Banking on India



Introduction

The unprecedented **India growth story.**

India's economic resurgence provides a robust long-term growth platform for a committed solution provider like BTL EPC

India is likely to grow into USD 5 trn economy in three years and towards a USD 10 trn economy during the next decade.

To make this a reality, more and larger industrial facilities will be needed.

The large industrial facilities of today could become the mid-sized of tomorrow and the small-sized of the next decade.

This evolution will put a premium on heavy engineering fabrication companies.

These companies will need to build their businesses faster to fabricate larger, advanced and different equipment.

This is seeding an unprecedented opportunity for these heavy engineering solution providing companies.

Companies like BTL EPC are attractively placed to capitalise on this unprecedented opportunity.

The big picture

Growth is everywhere in today's India.

This provides an
unprecedented opportunity to
the proactive and preponed

India. A
robust
USD
4.2 trn
economy

The world's fourth
largest economy

The world's consistently
fastest growing major
economy

The world's largest
population cluster

1 Infrastructure expansion

- National Infrastructure Pipeline (NIP) and PM Gati Shakti aim to invest over USD 1.5–2 Trillion by 2030.
- India's megaprojects comprise high-speed rail, Smart cities, mega ports, expressways, tunnels, metros, and multi-modal logistics hubs.

2 Energy transition and industrialisation

- India has targeted 500 GW of non-fossil fuel capacity by 2030.
- India is deepening capacity additions in its thermal, nuclear, and hydroelectric segments.

3 Manufacturing scale-up (Make in India / PLI schemes)

- Major push for electronics, defence, aerospace, automotive, shipbuilding, and capital goods.
- Heavy capex planned in sectors like steel, cement, petrochemicals, and fertilisers.

4 Transport and logistics megaprojects

- Dedicated Freight Corridors, Bharatmala, inland waterways, and airport/port expansion.
- India aims to become a global supply chain hub.

5 Defence and Aerospace

- India increasing indigenous production in shipbuilding, missiles, fighter jets, satellites.
- Public-private partnerships.

6 Upgradation of legacy assets

- India's older generation thermal plants, steel mills, and ports need modernisation.
- Focus on energy efficiency, emissions control, and automation.

a Heavy engineering needs

- Greater need for larger cranes, bridge-girders, tunnel boring machines, segment launchers and piling rigs.
- Need for modular fabrication systems and automated gantry systems for large pre-engineered structures.

b Heavy equipment needs

- Larger pressure vessels, boilers, wind turbine towers, electrolyzers, battery cell plant machinery, nuclear reactor components.
- Gigafactory-scale automation lines, robotic arms, and heavy-duty cleanroom equipment.

c Heavy equipment needs

- Larger rotary kilns, conveyors, boiler feed systems, metal presses, industrial furnaces, CNC machining centers, and robotic welding cells.
- Material handling systems like massive EOT (Electronic Overhead Travelling) cranes, gantries, and AGVs.

d Heavy equipment needs

- Heavy-duty conveyor systems, ship loaders, RTGs (Rubber Tired Gantry)/gantry cranes, container handling systems, and railway fabrication equipment.

e Heavy engineering needs

- Precision fabrication systems, metal forming machines, composite layup tools, engine test rigs, and missile handling equipment.

f Heavy equipment needs

- Flue gas desulphurisation (FGD) units, scrubbers, automated material handling systems, and IoT-enabled maintenance equipment.



Coal handling plant for Bangladesh India Friendship Power Company Limited (BIFPCL) Bangladesh Maitree

Overview

Sophistication is becoming necessary in **BTL EPC's heavy engineering solutions business.**

BTL EPC is responding to the growing national needs of scale, scope and sophistication

Demand for larger and more complex equipment:

As manufacturing plants grow, they require higher-capacity machinery, greater precision, and customised solutions. To meet these needs, heavy engineering suppliers must adopt more advanced design tools, materials technology, and fabrication techniques.

Tighter tolerances and higher standards:

Modern plants demand high reliability, energy efficiency, and digital integration (e.g., Industry 4.0 compatibility). Meeting these standards often requires more sophisticated engineering.

Automation: To remain competitive, engineering firms need to innovate

using automation, AI-assisted design, advanced simulations, etc.

Shorter lead times: Larger plants often work on tighter schedules. Meeting deadlines without compromising quality pushes suppliers to streamline operations and adopt smart manufacturing.

High captive manufacturing capability:

There is a greater priority in widening quality control across every leg of product manufacture than ever before. This makes it imperative for companies engaged in EPC projects to manufacture a significant part of the material in-house

Strong eco-system support: There is a growing need for large heavy engineering EPC companies to create supporting eco-systems of smaller specialised fabricators equipped to serve virtually as an extension of the larger partner's enterprise – deliver specialised products with speed

All round sustainability: There has been a visible shift for heavy engineering solution providers to engage in responsible and sustainable manufacturing operations – operations that pose no threat to the ecology, workers or neighbouring communities- with adequate documentation of improvements.

Why our sector is becoming increasingly complex

Demand for larger and more complex equipment

Tighter tolerances and higher standards

Automation

Shorter lead times

High captive manufacturing capability

Strong eco-system support

All-round sustainability



Our strategic **priority**

BTL EPC's heavy engineering fabrication facilities are evolving into modern showpieces.

BTL EPC has invested in process optimisation, digital integration, lean manufacturing, and workforce empowerment to build a modern heavy engineering solutions company

Process optimisation

Standardisation of workflows: BTL EPC defines and documents standard operating procedures for every stage-cutting, forming, welding, machining and inspection. This reduces variability and human error.

Process time reduction (Lean tools): The Company invested in Value Stream Mapping to identify and eliminate bottlenecks. It implemented 5S (Sort, Set, Shine, Standardise, Sustain) to maintain a clean, organised floor. It propagated Kaizen for small, continuous improvements led by shop floor teams.

Parallel processing: The Company adopts parallel processing by executing simultaneous operations, such as pre-machining in one section while welding occurs in another. Its modularised fabrication approach allows multiple units to advance at the same time.

Automation and technology deployment

CNC and robotic equipment: The Company has proactively invested in CNC cutting, robotic welding, and automated drilling to reduce cycle time and enhance repeatability.

Shop floor automation systems: The Company has deployed SCADA or manufacturing execution systems to monitor work orders, machine utilisation, and process times. The Company's real-time dashboards help supervisors respond with speed.

Automated testing: The Company integrated online or in-line testing to reduce rework and delays caused by manual inspection cycles.

Smart material handling

Efficient layout design: The Company redesigned the shop floor to minimise travel distance of materials and reduce handling. It uses U-shaped or linear layouts depending on product flow, enhancing space efficiency.

Mechanised movement systems: The Company invested in overhead cranes, AGVs, or conveyor systems to ensure the smooth flow of heavy components with minimal manual intervention.

Inventory and resource management

Just-in-Time (JIT) inventory: The Company progressively reduced clutter, minimised inventory cost, and liberated floor space. This helps the Company avoid the inefficiency of overproduction or excess waiting.

Digitised and inventory tracking: The Company invested in barcode/RFID-based systems integrated with ERP to track material use and availability in real time.

Skilled workforce utilisation

Cross-training: The Company is engaged in the flexible deployment of manpower based on workload, reducing idle time and bottlenecks.

Performance feedback systems: The Company introduced KPIs for each workstation and rewards high-performing teams. It uses dashboards to create healthy competition and real-time visibility.

AR/VR-based training: The Company reduces onboarding time and error rates in complex

assemblies or welding procedures through advanced training methods.

Quality control integration

Error-proofing (Poka-Yoke): The Company integrates jigs, fixtures, and checks to prevent errors before they occur. This helps avoid rework and wasted hours.

In-process quality checks: The Company implemented inspection gates after key operations instead of waiting until the final QC.

Digital tools and data analytics

Digital twin / simulation tools: The Company simulates shop floor operations to optimise layout and process flow. This helps identify weak spots in real-time production cycles.

Mobile devices for shop floor operators: The Company facilitates immediate feedback, reporting defects, accessing drawings or checklists on tablets/mobile devices.

Energy and waste management

Monitor utility consumption: The Company deployed advanced equipment to track compressed air, welding power, or furnace energy usage per unit. This helps identify leaks, overuse, or idling equipment.

Scrap and rework analysis: The Company records and analyses reasons for rework and scrap; this enhances insights into design or training.

Looking ahead

Why the BTL EPC business model is secure: Coal still has a future.

Overview

BTL EPC generates a sizable portion of its revenues linked to coal handling in thermal power plants and underground mines.

Even as there is a global movement away from coal towards renewable energy forms, India has sought an extension until 2070 to emerge as a Net Zero nation. This provides an opportunity-driven company like BTL EPC with multi-decade prospects of sustainable growth.

Thermal power provides over 70% of actual electricity generated, due to

intermittency of renewables. India's thermal power generation, especially coal-based power, will remain relevant across the years. It will act as a stabilising backbone to support the transition to renewable energy. It will be more focused on efficiency, emissions control, and reliability.

Base load stability: Thermal power will continue to provide stable, 24x7 base load to support the growing renewable energy mix.

Recent policy shifts: India plans to add 80 GW of coal capacity by 2032, mainly through supercritical and

ultra-supercritical units as well as the replacement of old inefficient plants. NTPC and other companies have revived stalled projects due to rising power demand.

Industrial and urban demand: Rapid urbanisation, electrification of transport, and industrial growth (cement, steel, and chemicals) will require more power.

The bottom line: Thermal power may decline in relative share, even as absolute capacity increases across the coming years.

Why the scenario for the coal sector is optimistic in India

Coal India Limited, India's largest producer, is deepening sectorial investments

45

New coal mines to be developed by CIL, Singareni Collieries and NLC India in five years

119

Projects being advanced by the Company

896

Million tonnes per year of capacity across 119 projects advanced by CIL

1,33,576

₹ Crore, sanctioned projects by CIL across five years (modern mining technologies and infrastructure)

42

% growth targeted by Ministry of Coal for Coal India Limited across five years

1,080

Million tonnes targeted by Ministry of Coal for Coal India Limited in 2025-26

1,533

Million tonnes targeted by Ministry of Coal for Coal India Limited by 2030-31

8

USD Billion proposed investment in coal-fired power plants near mines by Coal India Limited

Looking ahead

Why the BTL EPC business model is secure: Coal handling systems are moving to the next generation.



Coal handling plant for GSECL, Wanakbori, Gujarat

Overview

Coal handling systems in thermal power plants have become larger, more automated, and more environmentally responsible. They process greater volumes, improve reliability, reduce manpower, and comply with stricter environmental norms.

Scaling in capacity and size

Then: Early plants handled a few hundred tons per hour (TPH) of coal using belt conveyors, manual crushers, and basic stockpiles. Systems were designed for small generating units (<100 MW).

Now: Modern ultra-supercritical thermal plants (600 MW to 1,000+ MW) handle 2,000 to 4,000+ TPH of coal. Larger systems require multi-line conveyor galleries, dedicated marshalling yards, and fully automated control of bunkering and stockpile operations.

Automation and centralised control

Then: Manual or relay-based control systems with minimal automation. Operators manually adjusted feeder speeds, gates, and conveyor stops.

Now: PLC-SCADA systems manage the entire coal handling plant from

a central control room. Real-time data on belt speed, coal flow rate, crusher status, stacker-reclaimer position, etc. Predictive maintenance using condition monitoring (vibration sensors, thermography).

Material handling efficiency

Then: Use of simple wagon tippers or even side-discharge wagons for unloading. Manual operation of conveyors and limited redundancy.

Now: Fully automated tandem rotary wagon tippers, capable of handling 25–40 wagons/hour.

High-speed track hoppers with dust suppression and underground reclaim tunnels. Pipe conveyors used over long distances or in difficult terrains to reduce coal spillage and environmental impact.

Crushing and screening systems

Then: Basic single-stage crushers (like ring granulators). Oversized coal often bypassed or manually reduced.

Now: Two-stage crushing systems for precise size control (example, <25 mm). Roll crushers, hammer mills,

and impactors with adjustable output sizes. Vibrating screens integrated for oversize separation before bunkering or milling.

Stockyard management and mechanisation

Then: Open stockpiles managed using dozers and front-end loaders. Little or no automation in stacking or reclaiming.

Now: Stacker-reclaimers with luffing, slewing, and bucket wheel

mechanisms. Automated stockpile management with laser/radar level monitoring. Stockpiles planned to manage coal blending, reduce spontaneous combustion, and track age of coal.

Environmental and safety upgrades

Then: Dust, spillage, and water pollution were common. Limited fire detection, manual fire control systems.

Now: Dust suppression and extraction systems, including water mist, fog cannons, and bag filters.

Fire detection systems (infrared, temperature sensors, gas sensors) integrated into conveyors, bunkers, and hoppers. Enclosed conveyor galleries, belt conveyors with covers, and zero discharge concepts for ash and wash water.

Integration with railways and mines

Now: Some plants have integrated rail sidings and real-time train unloading systems connected to national logistics networks. Larger plants use merry-go-round systems or even

conveyor-based direct coal linkage from nearby mines. Weighbridges and RFID-based wagon tracking ensure real-time reconciliation of coal receipts and consumption.

Data analytics and optimisation

Coal flow is monitored digitally from arrival to burner. Fuel quality tracking (calorific value, moisture, ash content) is integrated with plant heat rate analysis. Some plants use AI/ML to predict bunker levels, coal flow bottlenecks, and optimise feeder sequences for minimal wear and maximum combustion efficiency.

ERP project BTL EPC D-365 (Microsoft D365-Business Central)

The BTL D365 Project marks a significant milestone for our organisation, as we transition to a more robust and efficient ERP platform to streamline our business processes and enhance overall productivity. Dynamics 365-Business Central enables to monitor and

control sales-order and purchasing process through automated workflows and dynamically updated inventory levels. It also helps to run the entire business and deliver greater results through predictive, AI-driven insights.